Round robins of modal agents in the Stag Hunt

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1 Introduction

Setting

In the Stag Hunt both players do best by hunting stag together; hunting stag alone pays nothing, and hare is a safe middle option. We write the payoffs as stag/stag a, stag against hare s, hare against stag t, hare/hare d, with $a > t > d > s$. The players are modal agents in the sense of Barasz et al. [1]: programs that decide by what can be proved about the opponent, with $p =$ "the opponent hunts stag against me" and $q =$ "the opponent hunts stag against HareBot". FairBot $\Box p$ hunts stag exactly when it can prove the opponent does. We take all 7,561 behaviourally distinct guarded agents of size at most 10 and let every agent play every agent, itself included, with equal weight (a round robin).

Result

- **FairBot never wins.** Normalising $a = 1$, $s = 0$, the triangle of strict Stag Hunts $0 < d < t < 1$ splits into exactly six regions, each won by a single agent, and FairBot wins none of them. The agent $A_3 = \Box(p \vee \Box^3\neg(p \vee q))$ wins on 44.6% of the area, including every game with payoffs $(4, 0, t, 2)$ for $2 < t < 4$ and the game $(3, 0, 2, 1)$. At $(4, 0, 3, 2)$ it scores 23,411 against FairBot's 21,719.

- **Why it beats FairBot.** A_3 reaches stag/stag with 4,444 of the 4,459 agents that hunt stag against StagBot and with none of the others; it gains on the first group and loses on the second, for a net gain of 1,692 points over FairBot at $(4, 0, 3, 2)$.

- **Families behind the winners.** For every k, $A_k = \Box(p \vee \Box^k\neg(p \vee q))$ hunts stag against HareBot (and is left alone there), coordinates with StagBot, with every A_m and with every $F_j = \Box(\Box^j\bot \vee p)$ (FairBot over a stronger system); the lenient FairBots $R_k = \Box(p \vee \Box^k p)$ are safe and coordinate with $\neg\Box^j\bot$ exactly when $j \leq k$.

- **The winner depends on the size bound.** In the same census the score of R_k rises strictly with k up to $k = 8$ and that of A_k up to $k = 5$, so longer members of both families score higher than the census winners. The specific winners above are therefore artefacts of the size bound; what

holds across the families is that lenient agents beat FairBot in this round robin.

Why it matters

Round robins are a natural way to compare programs that read each other's code, and FairBot is often taken as the reference cooperative agent. In the Stag Hunt, where cooperating with a defector costs less than in the prisoner's dilemma, agents that are more willing to hunt stag do better than FairBot against a broad population, and exact payoff breakpoints show where each agent takes over.

Proved vs checked

The laws for A_k and R_k are proved for all indices in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [29], cited, not formalised (see also [12]). Scores, regions and breakpoints are exact rational computations over the stated census.

Novelty statement

We believe this result is new. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire et al., arXiv 1401.5577.
- LessWrong / Alignment Forum wiki "Modal combat" and "Is there only one FairBot?" (2026).
- Mennen, "Modal combat for games other than the prisoner's dilemma" (2017).
- Todorov, ten Napel & Müller, "Parametric Open Source Games", arXiv 2606.27068.
- Tennenholtz (2004), program equilibrium (background).
- Web searches for modal agents in the Stag Hunt and for tournaments of modal agents (2026-10-21); none means not found in this search.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly checked by scripts that run Lean proofs (claims K8, K3) and partly checked by script (claims K15, K6, K14, K5, K7, K4, K17, K18); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K15

Normalising $a = 1, s = 0$, the strict Stag Hunt triangle $0 < d < t < 1$ splits into exactly 6 census (size ≤ 10) round-robin winner regions, each one agent; A_3 wins on 44.6% of the area and $R_6 = \Box(p \vee \Box^6 p)$ near $t, d \to 1$; FairBot nowhere.

Status: derived ů author check only.

Verified.

- `checks/K15-c046_sh_triangle.py`: 7561 classes re-enumerated (own engine); K13/K14 values; 3309 score planes; exact Fraction clipping gives 6 winner polygons with total area exactly 1/2, one agent each, exact areas and A3 vertices; unique exact argmax at 7 interior points; 6 GLS self-certificates — pass

Evidence.

- reasoning: Round-robin argmax is invariant under positive affine payoff maps, so (a,s,t,d) reduces to (1,0,t,d); scores are affine in (t,d), so winner regions are convex polygons.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K6

In the 7,561-class guarded-language census of size ≤ 10, $A_3 = \Box(p \vee \Box^3 \neg(p \vee q))$ uniquely maximizes round-robin score $21950 + 487t$ for every $2 < t < 4$, with payoffs $(SS, SH, HS, HH) = (4, 0, t, 2)$.

Status: derived ů author check only.

Verified.

- `checks/K6-stag_payoffs.py`: Pinned solver source; exact census counts; all ordered cells solved; GLS certificates for A3/self, A3/HareBot, A3/FairBot and FairBot/self; exact outcome counts, transitions, and upper envelopes for both payoff slices. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K14

At $(SS, SH, HS, HH) = (3, 0, 2, 1)$, $A_3 = \Box(p \vee \Box^3 \neg(p \vee q))$ uniquely maximizes the equally weighted round robin over the 7561 guarded size-≤ 10 census classes, scoring 16393 versus FairBot's 14158.

Status: derived ů author check only.

Verified.

- `checks/K14-todorov_safety.py`: SHA256-pinned solver; all 7561-class cells re-solved; unique winner A3 at (3,0,2,1), score 16393 versus FairBot 14158; 7561 winner cells independently re-solved; 4 GLS certificates; 1693 census-safe classes, 517 self-stag. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K5

For $(SS, SH, HS, HH) = (4, 0, 3, h)$ and $0 < h < 4$, the size-≤ 10 modal census has six winner intervals, separated by 116/1423, 15/22, 2537/1001, 4079/1071, and 463/116.

Status: derived ů author check only.

Verified.

- `checks/K6-stag_payoffs.py`: Pinned solver source; exact census counts; all ordered cells solved; GLS certificates for A3/self, A3/HareBot, A3/FairBot and FairBot/self; exact outcome counts, transitions, and upper envelopes for both payoff slices. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K7

In L up to size 10 with Stag Hunt $(4, 0, 3, 2)$, $A_3 = \Box(p \vee \Box^3\neg(p \vee q))$ ends stag/stag with exactly 4444 of the 4459 opponents that hunt stag vs StagBot and with none of the other 3102; vs FairBot it gains +3848 on the first group and loses 2156 on the second.

Status: derived ů author check only.

Verified.

- `checks/K7-c046_xreview_k4k6.py`: 7561 classes; A3/FB counts and K4 transitions; K5/K6 envelopes; StagBot split 4459/3102, A3 SS = 4444 all in first group, 15 HS exceptions of size >=9 with []^4, gains +3848/-2156; 60488 cells re-solved independently; 4 GLS certificates — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K4

At Stag Hunt payoffs $(4, 0, 3, 2)$ in the size-≤ 10 census, A_3 gains 1,692 points over FairBot: 1,643 opponents change $HH \to SS$ and 562 change $HS \to SS$, offset by other outcome changes.

Status: derived ů author check only.

Verified.

- `checks/K6-stag_payoffs.py`: Pinned solver source; exact census counts; all ordered cells solved; GLS certificates for A3/self, A3/HareBot, A3/FairBot and FairBot/self; exact outcome counts, transitions, and upper envelopes for both payoff slices. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K8

For all $k, j, m \geq 0$, A_k plays stag/hare vs HareBot and stag/stag vs StagBot, A_m, and $F_j = \Box(\Box^j \bot \vee p)$. Against $PB_j = \Box p \wedge \Box(\Box^{j+1} \bot \vee \neg q)$, it plays stag/hare for $k = 0$ and hare/hare for $k \geq 1$.

Status: derived ů author check only.

Verified.

- `checks/K8-ak_laws.py`: Seven core Lean theorems for all indices and arbitrary solution traces; no sorryAx; incorrect strict bound rejected; 261 independent height cells; eight GLS fixed-point certificates; pinned prior solver source. — pass (Lean-proved, no sorryAx)
- `checks/K8-ak_laws.py`: 11 core Lean theorem reports, no sorryAx; faithful PB guard equivalent to original P_(j+1) when q=true; all-index A/PB outcomes; false mutation rejected; 342 height cells; 12 GLS certificates. — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K3

For all $k \geq 0$, $R_k = \Box(p \vee \Box^k p)$ (R_0 = FairBot) never ends hunting stag against an opponent that ends hunting hare, and in the Stag Hunt ends stag/stag with $E_j = \neg\Box^j \bot$ iff $j \leq k$ (else hare vs stag) and with $N_n = \Box(\Box^n \bot \vee \neg q)$ iff $n > k$ (else hare/hare).

Status: derived ů author check only.

Verified.

- `checks/K3-c046_lenient_laws.py`: Lean (17 theorems, no sorryAx): rk_safe, rr_pair, rk_db, rk_E (stag/stag iff j<=k), rk_N (stag/stag iff n>k, else hare/hare), modal bridges; false variant rejected; height tables k<=7, j,n<=8; 14 GLS certificates — pass (Lean-proved, no sorryAx)
- `checks/K3-c046_relabel_gk.py`: Lean re-proves rk_E, rk_N, rk_Z, rk_db for all indices alongside gk_E, gk_N (same E_j/N_n laws for G_k); no sorryAx; false bound rejected; 588 height cells; 10 GLS certificates — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Height semantics on the GL chain (box holds at h iff its argument holds at all g\<h). Game tails equal standard-model actions; Lean proves the laws for every index, given any traces solving the height equations.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K17

Against the 7561 size-
le10 census agents, $R_k =$
$Box($p
lor
$Box^k p$) has round-robin score strictly increasing in k for 0
lek
le8 and constant for 8
lek
le12, for all payoffs with $a > t, a > d$; so R_6's K15 region reflects the size bound (size $k + 4$).

Status: derived ů author check only.

Verified.

- `checks/K17-c046_rk_artifact.py`: 7561 classes re-enumerated; R_k counts k=0..12 (machine solver = height solver for k<=6), SH=0; SS up, HS/HH non-increasing, strict until k=8 then constant; 200 exact random payoffs: score strictly increasing to k=8, R_8-R_6=4(a-t); R_6->R_7 differs on ~[]^7F,~[]^7~p,~[]^7~q, R_7->R_8 on ~[]^8F; 4 GLS certificates — pass

Evidence.

- reasoning: SH=0 for all k and counts sum to 7561, so the score difference is -dHS(a-t)-dHH(a-d), positive whenever dHS,dHH\<=0 with one strict and a\>t,a\>d.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K18

Against the 7561 size-≤ 10 census agents, $A_k = \Box(p \vee \Box^k \neg(p \vee q))$ (size $k + 7$) has 543 SH and 2087 HH for $1 \leq k \leq 10$; its score rises strictly for $k \leq 5$ (by $\Delta SS(a - t)$) and is constant after, so $A_5 - A_3 = 15(a - t)$ whenever $a > t$.

Status: derived ů author check only.

Verified.

- `checks/K18-c046_ak_artifact.py`: 7561 classes re-enumerated; A_k counts k=1..10 (machine = height solver for k<=3, A_4.. not census members); SH 543, HH 2087 fixed; only HS->SS for k>=2; 200 exact random payoffs: strict increase to k=5, constant to k=10, A_5-A_3=15(a-t); A_5 SS set = 4459 stag-vs-StagBot agents; 12 then 3 listed flips; 5 GLS certificates — pass

Evidence.

- reasoning: SH and HH counts are constant and counts sum to 7561, so the score difference is dSS\a + dHS\t = dSS(a-t) \> 0 when dSS\>0 and a\>t.

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261021-0517`, claims K15, K6, K14, K5, K7, K4, K8, K3, K17, K18, run-folder commit `580565ffd3b3f1a8ca3ffa0407c949c8d724865c`, published 2026-10-21.

3 Discussion

Round-robin results depend on the opponent population: here all small formulas with equal weight, which over-represents some behaviours. The scores are aggregate quantities, not equilibrium thresholds; they are not comparable to the learning-dynamics threshold of Todorov et al. for parametric (non-modal) agents [30]. Open: whether the scores of A_k and R_k keep their order in larger censuses, and which agent wins in the limit.

4 How to verify

Each finding has its check scripts in `claims/staghunt-round-robin/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[3] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[4] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[5] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[6] Christoph Benzmüller. Proofs without nominals: Gödel's ontological argument, its shallow embedding, and the open questions of the monatshefte notes. arXiv:2609.36279v4, 2026.

[7] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[8] Christoph Benzmüller and Lawrence C. Paulson. Quantified multimodal logics in simple type theory. *Logica Universalis*, 7(1), 2013.

[9] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[10] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[11] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[12] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[13] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[14] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[15] Kit Fine. Propositional quantifiers in modal logic. *Theoria*, 36(3), 1970.

[16] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[17] Murdoch J. Gabbay and Dominic P. Mulligan. Nominal Henkin semantics: simply-typed lambda-calculus models in nominal sets. arXiv:1111.0089, 2011.

[18] Gram_Stone. Modal chicken. LessWrong, 2015.

[19] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[20] Wesley H. Holliday. A note on algebraic semantics for S5 with propositional quantifiers. *Notre Dame Journal of Formal Logic*, 60(2), 2019.

[21] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[22] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[23] Mladen Mika and Jakob Nordström. A generalized method for proving polynomial calculus degree lower bounds. In *30th Conference on Computational Complexity (CCC 2015)*, volume 33 of *LIPIcs*, pages 467–487, 2015.

[24] Reinhard Muskens. Intensional models for the theory of types. *Journal of Symbolic Logic*, 72(1), 2007.

[25] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[26] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[27] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[28] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.

[29] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[30] Aleksandar Todorov, Jesse ten Napel, and Alexander Müller. Parametric open source games. arXiv:2606.27068, 2026.

[31] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.